A3.3 Relationships between the channels

The AXI protocol requires the following relationships to be maintained:

- a write response must always follow the last write transfer in the write transaction of which it is a part
- read data must always follow the address to which the data relates
- channel handshakes must conform to the dependencies defined in *Dependencies between channel handshake signals*.

Otherwise, the protocol does not define any relationship between the channels.

This means, for example, that the write data can appear at an interface before the write address for the transaction. This can occur if the write address channel contains more register stages than the write data channel. Similarly, the write data might appear in the same cycle as the address.

Note

When the interconnect is required to determine the destination address space or slave space, it must realign the address and write data.This is required to assure that the write data is signaled as valid only to the slave for which it is destined.

A3.3.1 Dependencies between channel handshake signals

To prevent a deadlock situation, the dependency rules that exist between the handshake signals must be observed.

As summarized in *Channel signaling requirements* on page A3-38, in any transaction:

- the **VALID** signal of the AXI interface sending information must not be dependent on the **READY** signal of the AXI interface receiving that information
- an AXI interface that is receiving information can wait until it detects a **VALID** signal before it asserts its corresponding **READY** signal.

Note

While it is acceptable to wait for **VALID** to be asserted before asserting **READY**, it is also acceptable to assert **READY** before detecting the corresponding **VALID**. This can result in a more efficient design.

In addition, there are dependencies between the handshake signals on different channels, and AXI4 defines an additional write response dependency. The following subsections define these dependencies:

- *Read transaction dependencies* on page A3-41
- *Write transaction dependencies* on page A3-41
- *AXI4 write response dependency* on page A3-42.

In the dependency diagrams:

- single-headed arrows point to signals that can be asserted before or after the signal at the start of the arrow
- double-headed arrows point to signals that must be asserted only after assertion of the signal at the start of the arrow.

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Read transaction dependencies

Figure A3-5 shows the read transaction handshake signal dependencies, and shows that, in a read transaction:

- the master must not wait for the slave to assert **ARREADY** before asserting **ARVALID**
- the slave can wait for **ARVALID** to be asserted before it asserts **ARREADY**
- the slave can assert **ARREADY** before **ARVALID** is asserted
- the slave must wait for both **ARVALID** and **ARREADY** to be asserted before it asserts **RVALID** to indicate that valid data is available
- the slave must not wait for the master to assert **RREADY** before asserting **RVALID**
- the master can wait for **RVALID** to be asserted before it asserts **RREADY**
- the master can assert **RREADY** before **RVALID** is asserted.

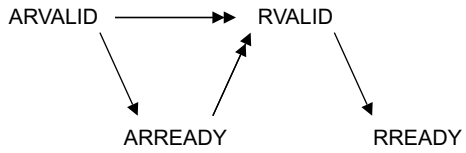


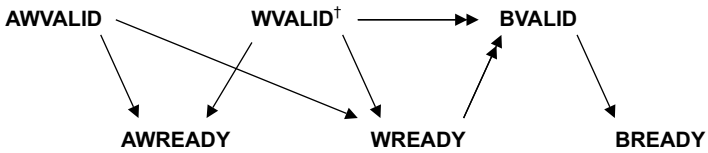
Figure A3-5 Read transaction handshake dependencies

Write transaction dependencies

Figure A3-6 shows the write transaction handshake signal dependencies, and shows that in a write transaction:

- the master must not wait for the slave to assert **AWREADY** or **WREADY** before asserting **AWVALID** or **WVALID**
- the slave can wait for **AWVALID** or **WVALID**, or both before asserting **AWREADY**
- the slave can assert **AWREADY** before **AWVALID** or **WVALID**, or both, are asserted
- the slave can wait for **AWVALID** or **WVALID**, or both, before asserting **WREADY**
- the slave can assert **WREADY** before **AWVALID** or **WVALID**, or both, are asserted
- the slave must wait for both **WVALID** and **WREADY** to be asserted before asserting **BVALID**

  the slave must also wait for **WLAST** to be asserted before asserting **BVALID**, because the write response, **BRESP**, must be signaled only after the last data transfer of a write transaction
- the slave must not wait for the master to assert **BREADY** before asserting **BVALID**
- the master can wait for **BVALID** before asserting **BREADY**
- the master can assert **BREADY** before **BVALID** is asserted.



† Dependencies on the assertion of **WVALID** also require the assertion of **WLAST**

Figure A3-6 Write transaction handshake dependencies

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Caution

The dependency rules must be observed to prevent a deadlock condition. For example, a master must not wait for **AWREADY** to be asserted before driving **WVALID**. A deadlock condition can occur if the slave is waiting for **WVALID** before asserting **AWREADY**.

AXI4 write response dependency

AXI4 defines an additional AXI4 slave write response dependency.

Note

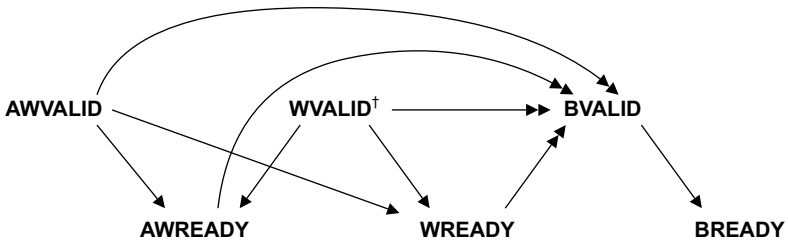
- this additional dependency reflects the expected use in AXI3, because it is not expected that any components would accept all write data and provide a write response before the address is accepted
- by issuing a write response, the slave takes responsibility for hazard checking the write transaction against all subsequent transactions.

Figure A3-7 shows all the AXI4 required slave write response handshake dependencies. The single-headed arrows point to signals that can be asserted before or after the previous signal is asserted. Double-headed arrows point to signals that must be asserted only after assertion of the previous signal.

Figure A3-7 summarizes the AXI4 slave write response handshake dependencies. These dependencies are:

- the master must not wait for the slave to assert **AWREADY** or **WREADY** before asserting **AWVALID** or **WVALID**
- the slave can wait for **AWVALID** or **WVALID**, or both, before asserting **AWREADY**
- the slave can assert **AWREADY** before **AWVALID** or **WVALID**, or both, are asserted
- the slave can wait for **AWVALID** or **WVALID**, or both, before asserting **WREADY**
- the slave can assert **WREADY** before **AWVALID** or **WVALID**, or both, are asserted
- the slave must wait for **AWVALID**, **AWREADY**, **WVALID**, and **WREADY** to be asserted before asserting **BVALID**

  the slave must also wait for **WLAST** to be asserted before asserting **BVALID** because the write response, **BRESP** must be signaled only after the last data transfer of a write transaction
- the slave must not wait for the master to assert **BREADY** before asserting **BVALID**
- the master can wait for **BVALID** before asserting **BREADY**
- the master can assert **BREADY** before **BVALID** is asserted.



† Dependencies on the assertion of **WVALID** also require the assertion of **WLAST**

Figure A3-7 Slave write response handshake dependencies

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A3.3.2 Legacy considerations

The additional dependency described in *AXI4 write response dependency* on page A3-42 means that an AXI3 slave that accepts all write data and provides a write response before accepting the address is not compliant with AXI4. Converting an AXI3 legacy slave to AXI4 requires the addition of a wrapper that ensures a returning write response is not provided until the appropriate address has been accepted by the slave.

Note

This specification strongly recommends that any new AXI3 slave implementation includes this additional dependency.

Any AXI3 master complies with the AXI4 write response requirements.

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